

Question Paper

COURSE : Machine Learning (CS F464)

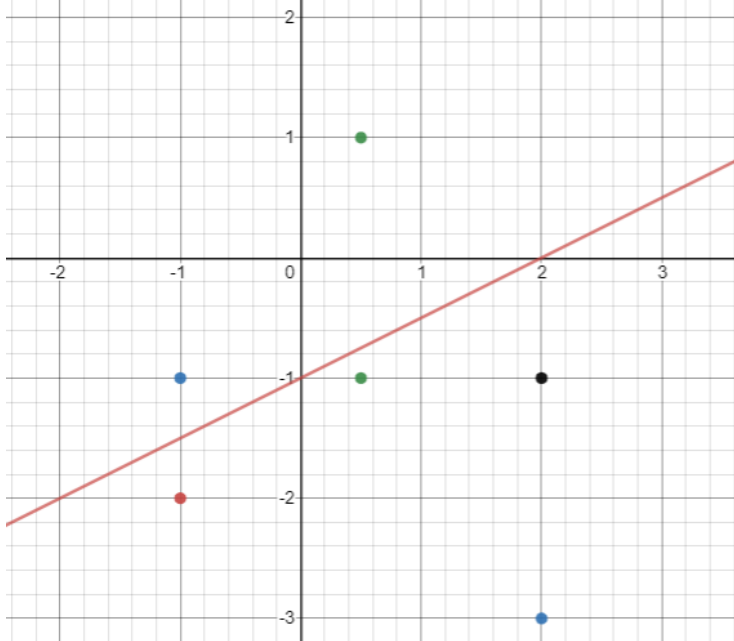
COMPONENT : Test 1 CLOSED BOOK

WEIGHTAGE : 20% (20 Marks)

DATE : 9-March-2021, T2

DURATION : 50 Minutes (8.30 AM - 9.20 AM (GST))

Note: Please write down any assumptions that you make.

Q. 1	Mention the types of discriminant classifiers and write a one line note about them. Can they be applied to classify multi class models?	2 M										
Q. 2	Let's plot the order of polynomial on the X axis and the RMS error on the Y axis for the training data and test data. Draw a justifiable graph and discuss over how to choose the optimal number of polynomial terms. Discuss about the over fitting and underfitting phenomenon.	2 M										
Q. 3	Few observations are available for Class 1 and Class 2 models <table><tr><td>Class 1</td><td>2</td><td>4</td><td>6</td><td>4</td></tr><tr><td>Class 2</td><td>10</td><td>12</td><td>8</td><td>10</td></tr></table> Assume that we are building a univariate gaussian classifier and classify to which class the point with value 8 would belong to. Assume both the class have equal prior probabilities. HINT: Apply log probability ratio	Class 1	2	4	6	4	Class 2	10	12	8	10	2 M
Class 1	2	4	6	4								
Class 2	10	12	8	10								
Q. 4	Given a decision boundary line: $2x+y=4$. Given the point: (2,3), (0.5,1) and (2,1). Apply formulas of discriminant function and calculate the projected distance of the points from the decision boundary. Find to which side of the decision boundary the point belongs to.	2 M										
Q. 5	Given the line and 6 points.  Calculate the equation of the line and the coordinates of the points. Assume the line to be the predicted line of Linear Regression model and the points to be the target points. Write the error formulae of Linear Regression and Calculate the Error value.	3 M										

Q. 6

Given the following set of training examples.

4 M

A	B	C	CLASS LABEL 1	CLAS LABEL 2
1	GREEN	20	TRUE	FALSE
2	GREEN	30	FALSE	FALSE
3	GREEN	20	TRUE	FALSE
4	YELLOW	30	FALSE	TRUE
5	YELLOW	20	TRUE	TRUE
6	YELLOW	30	FALSE	TRUE

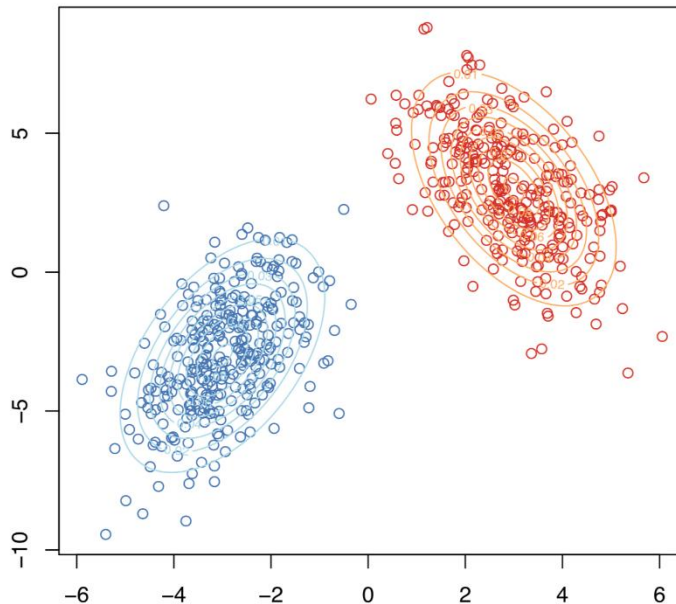
1. Calculate the information gain for attributes B and C with respect to Class label 1 and Class Label 2.
2. Of the attributes B and C, which attribute can be opted as splitting attribute with respect to Class label 1.
3. Of the attributes B and C, which attribute can be opted as splitting attribute with respect to Class label 2.

Q. 7

Given the two class of data points. Consider projecting the D-dimensional input vector into one dimension and draw the decision boundary such that the two class of points have to be separated as wider as possible.

5 M

Which method do you adopt for the same? Derive the method and conclude the factors that contributes to separating the two class of points.



***** ALL THE BEST *****

Question Paper

COURSE : Machine Learning (CS F464)

COMPONENT : Test 2 OPEN BOOK

WEIGHTAGE : 20% (20 Marks)

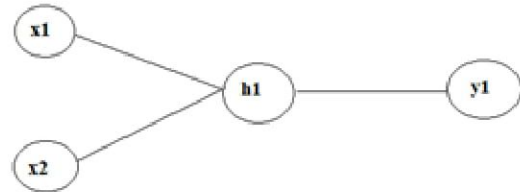
DATE : 18-April-2021, Su2

DURATION : 50 Minutes (8.30 AM - 9.20 AM (GST))

Note: Please write down any assumptions that you make.

Q. 1

Given:
Input@x1=0.5
Input@x2=0.3
wt(x1,h1)=0.1
wt(x2,h1)=0.2
wt(h1,y1)=0.3
bias@h1=0.1
bias@h2=0.2
Target value @y1=0.9
Find the Error at y1 via link (h1, y1).

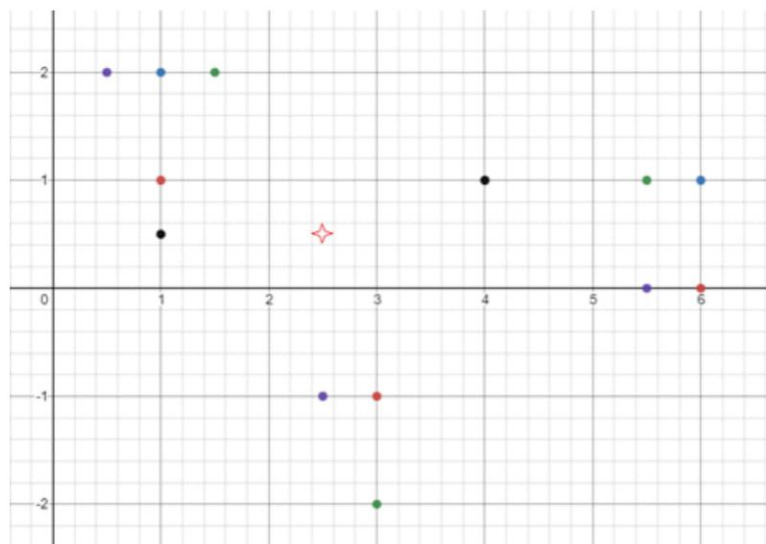


2

Q. 2

Class 1	Class 2	Class 3
1→(1, 1)	6→(6, 0)	11→(3, -1)
2→(1, 2)	7→(6, 1)	12→(3, -2)
3→(1.5, 2)	8→(5.5, 1)	13→(2.5, -1)
4→(0.5, 2)	9→(5.5, 0)	
5→(1, 0.5)	10→(4, 1)	

Given 13 points grouped in the three classes. Apply k-NN with k value as 6 and allocate the point(2.5, 0.5) to the suitable class. Choose the nearest 6 points from the graph itself.



3

Q. 3

There are 4 laboratories whose probability of prediction is given by {0.3, 0.4, 0.2, 0.1}.
Person-A tests a sample-1 in all the 4 laboratories. The laboratory 2 and 4 tests the sample-1 to be malignant.
Person-B tests a sample-2 in all the 4 laboratories. The laboratory 1 and 2 tests the sample-2 to be tumor.
Predict whether sample-1 is malignant or tumor. Also predict for sample-2.

5

Q. 4

Think over the geometric intuition behind SVM. Try creating an optimal hyperplane that separates the two class of points as wider as possible. Derive the W vector.

Find the mid boundary equation.

Class 1

Pt 1 $\rightarrow (1.5, 2)$

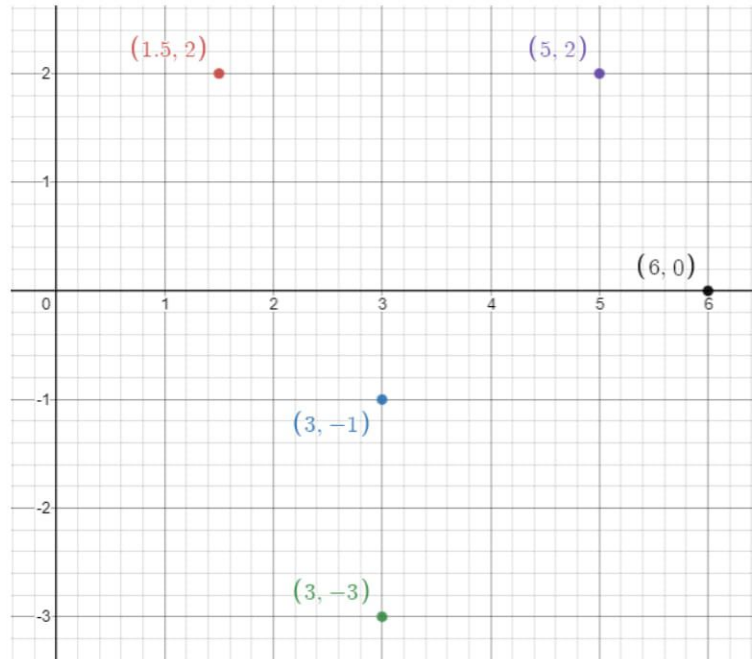
Pt 2 $\rightarrow (3, -1)$

Pt 3 $\rightarrow (3, -3)$

Class 2

Pt 4 $\rightarrow (5, 2)$

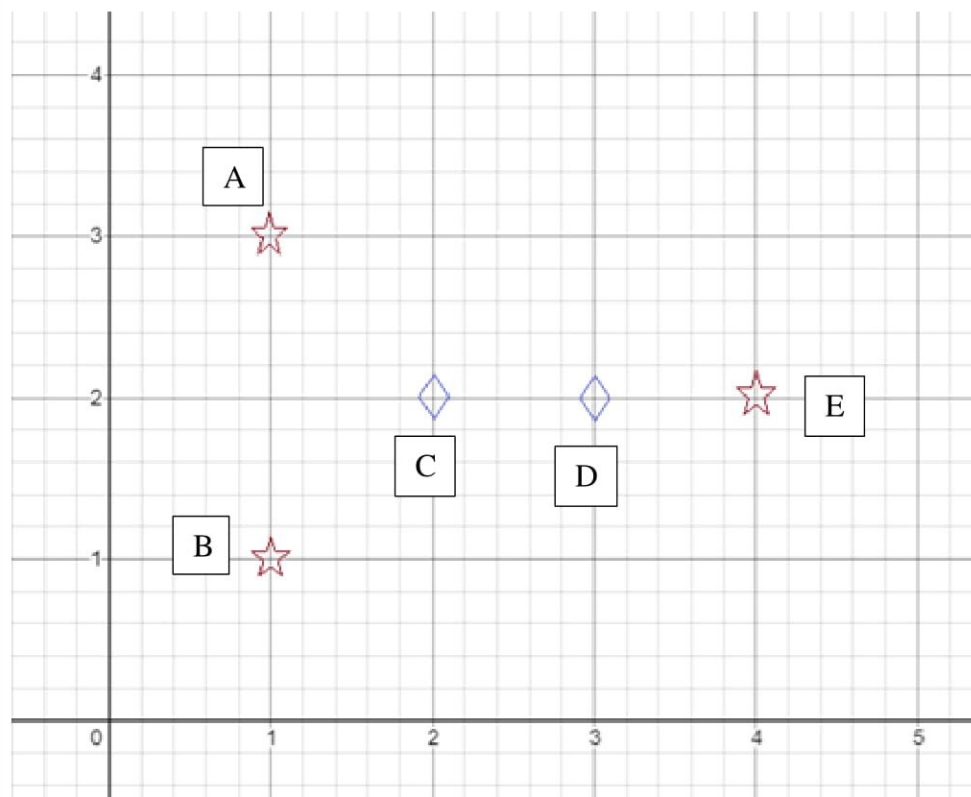
Pt 5 $\rightarrow (6, 0)$

**5****Q. 5**

Given the two class of points. Construct a sequential learner by applying an iterative approach to learn from the weak classifiers and build a strong classifier which would finitely classify the two class of points.

Solve two iterations in steps and calculate the voting power of each of the two weak classifiers until all the points are classified finitely.

NOTE: Choose only the optimal weak classifiers.

**5**

***** ALL THE BEST *****

**DUBAI INTERNATIONAL ACADEMIC CITY, DUBAI
SECOND SEMESTER 2020 – 2021
DEPARTMENT OF COMPUTER SCIENCE**

**Questions: 5
Pages: 3**

COURSE : Machine Learning (CS F464)

COMPONENT : Comprehensive QP-OPEN BOOK
SECTION 1

WEIGHTAGE : 40% (40 Marks)

SECTION 1: 20 Marks

DATE : 01-June-2021, AN

DURATION : 80 Minutes (12.30 PM – 01:50 PM (GST))

Note: Please write down any assumptions that you make.

Q. 1

A. A store orders for books every month based on the demand. The sale of the book depends on the number of customers visiting the shop. The table given below shows the customer count and books sold in the year 2020

Month	Customer count	Books sold
1	6	1
2	8	7
3	5	4
4	9	5
5	8	6
6	8	8
7	6	4
8	7	6
9	6	3
10	3	1
11	5	2
12	8	7

- For the year 2020, apply the linear prediction model using the equation $y=x-a$ by applying values for 'a' in two iterations as '3' and '2'.
- Build a table and mention the calculated predicted books sold for both the cases
- Calculate the error values for both the case and identify the ideal linear equation
- With the chosen ideal linear equation, predict the books sold for the month Jan, 2021, if the customer count 11.

B. The details of the same store for the past 6 yrs is given in the table below and includes the books sold for the month of Jan.

Analyze the values in the below table and choose your own suitable linear equation that owns minimal error.

With the chosen linear equation, predict the values sold for the month of Jan 2021, if the customer count 11.

SNO	Customer Count	Books sold
Jan 2015	6	7
Jan 2016	3	5
Jan 2017	2	3
Jan 2018	4	5
Jan 2019	4	5
Jan 2020	5	6

Is the prediction results for the month of Jan 2021 is same or different in both the cases.

**4 M
(3+1)**

Q. 2

Given the table with attributes shape, size, color and the property as class label.

4 M

1. Calculate the entropy value.
2. Calculate the information Gain for the attributes (shape and color) given in the table and among them chose the attribute with the best information gain as a split attribute for the first level of classification.

Shape	Size	Color	Property → Class Label
Oval	Small	Black	Y
Oval	Large	Blue	N
Oval	Small	Black	Y
Round	Large	Blue	N
Round	Small	Black	Y
Round	Large	Black	Y
Round	Small	Black	Y
Square	Large	Blue	N
Square	Small	Blue	N
Square	Large	Blue	N
Square	Small	Black	Y

Q. 3

Each of 13 soldiers with ID Z,Y,X,W,V,U,T,S,R,Q,P,O,N would like to reach the Goal state. The reward to reach the goal state from states 'p' and 'n' is 10000.

4 M (2+2)

1. Calculate the values for every state using a discount factor of 0.95
2. Apply the Q-learning formula – discounted reward (discount factor =0.95) and find the reward of moving from every state to every other state

Z	Y	X	
W	V	U	T
S	R	Q	P
	O	N	GOAL

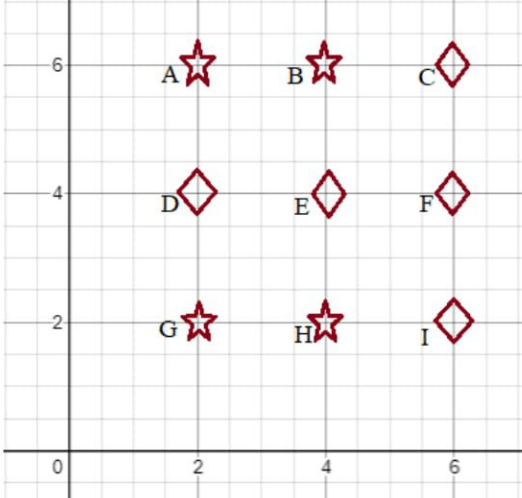
Q. 4	Given below the datapoints. <table><tr><th>Points</th><th>Co-ordinates</th></tr><tr><td>1</td><td>1,-1</td></tr><tr><td>2</td><td>0,4</td></tr><tr><td>3</td><td>0,2</td></tr><tr><td>4</td><td>1,-2</td></tr><tr><td>5</td><td>0,-1</td></tr><tr><td>6</td><td>1,1</td></tr><tr><td>7</td><td>0,1</td></tr><tr><td>8</td><td>-1,1</td></tr></table> 1. Apply k-Means clustering on them with k value as 2. Choose C1 (0,1) and C2 (0,-1) as centroid value in the first iteration and identify to which cluster the points belong to? 2. Just guess → If you iterate for the second time will there any change seen in the points allocated to clusters?	Points	Co-ordinates	1	1,-1	2	0,4	3	0,2	4	1,-2	5	0,-1	6	1,1	7	0,1	8	-1,1	4 M (3+1)												
Points	Co-ordinates																															
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5	0,-1																															
6	1,1																															
7	0,1																															
8	-1,1																															
Q. 5	Climate Vs umbrella Assume the probability of day to be rainy day, sunny day or cloudy day is same on the first day. In general, the probability of shift of climate from one day to another is given in table below: <table><tr><th>Day 1</th><th>Day 2</th><th>Probability value</th></tr><tr><td>Rainy</td><td>Rainy</td><td>0.5</td></tr><tr><td>Rainy</td><td>Sunny</td><td>0.3</td></tr><tr><td>Rainy</td><td>Cloudy</td><td>0.2</td></tr><tr><td>Sunny</td><td>Rainy</td><td>0.4</td></tr><tr><td>Sunny</td><td>Sunny</td><td>0.4</td></tr><tr><td>Sunny</td><td>Cloudy</td><td>0.2</td></tr><tr><td>Cloudy</td><td>Rainy</td><td>0.3</td></tr><tr><td>Cloudy</td><td>Sunny</td><td>0.3</td></tr><tr><td>Cloudy</td><td>Cloudy</td><td>0.4</td></tr></table> In general, on a rainy day the probability of me taking an umbrella is given by 0.6 and on a sunny and cloudy day the probability of me taking an umbrella is 0.4 and 0.7 respectively. 1. a. Find the probability of raining on the second day b. Find the probability of sunny on the second day 2. On the second day, find the probability that I have taken my umbrella and it rains	Day 1	Day 2	Probability value	Rainy	Rainy	0.5	Rainy	Sunny	0.3	Rainy	Cloudy	0.2	Sunny	Rainy	0.4	Sunny	Sunny	0.4	Sunny	Cloudy	0.2	Cloudy	Rainy	0.3	Cloudy	Sunny	0.3	Cloudy	Cloudy	0.4	4 M (2+2)
Day 1	Day 2	Probability value																														
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***** ALL THE BEST *****

Question Paper

COURSE : Machine Learning (CS F464)	
COMPONENT : Compre QP-OPEN BOOK SECTION 2	WEIGHTAGE : 40% (40 Marks) SECTION 2: 20 Marks
DATE : 01-June-2021, AN	DURATION : 85 Minutes (02.00 PM - 03.25 PM (GST))
Note: Please write down any assumptions that you make.	

Q. 1	Apply Linear Discriminant Analysis (LDA) dimensionality reduction approach on the given points and calculate the new projected values of the data points. After the execution of LDA, find the discriminant values of all the points towards cluster-1 only. <table border="1"> <thead> <tr> <th>X-coordinate</th><th>Y-coordinate</th><th>Class</th></tr> </thead> <tbody> <tr><td>1</td><td>1</td><td>1</td></tr> <tr><td>1</td><td>3</td><td>1</td></tr> <tr><td>1</td><td>2</td><td>1</td></tr> <tr><td>-1</td><td>1</td><td>2</td></tr> <tr><td>-1</td><td>2</td><td>2</td></tr> <tr><td>-1</td><td>3</td><td>2</td></tr> </tbody> </table>	X-coordinate	Y-coordinate	Class	1	1	1	1	3	1	1	2	1	-1	1	2	-1	2	2	-1	3	2	8 M
X-coordinate	Y-coordinate	Class																					
1	1	1																					
1	3	1																					
1	2	1																					
-1	1	2																					
-1	2	2																					
-1	3	2																					
Q. 2	Given a sample to a hypothesis, it can be classified into (yes or no). Let say there are 5 hypothesis which predicts a given sample into one of the two labels: yes/no. The probability of hypothesis 1, hypothesis 2, hypothesis 3, hypothesis 4, hypothesis 5 producing the right results for any given dataset is 0.5, 0.6, 0.4, 0.7, 0.8 respectively. Given a sample, the hypothesis 1, 3, 5 predicts it as yes whereas the hypothesis 2, 4 predicts it as no. Find whether the sample belong to class yes or no.	3 M																					
Q. 3	Being a professional who works on a very large unlabeled dataset. The task is to label the unlabeled dataset and train the model. Labelling a single sample might cost a huge value and hence it becomes a critical task to choose the right sample, so that the model gets trained with minimal cost. Assume you have chosen few samples and identified their probability values of belonging to two classes. 1. Calculate the smallest-margin uncertainty value and ratio of confidence uncertainty value for all the samples. 2. Identify which sample you would prefer to label using smallest-margin uncertainty sampling and ratio of confidence uncertainty sampling.	3 M (2 +1)																					

	<table> <tr> <th>Sample</th><th>P(C1/Sample)</th><th>P(C2/sample)</th></tr> <tr> <td>1</td><td>0.85</td><td>0.57</td></tr> <tr> <td>2</td><td>0.21</td><td>0.99</td></tr> <tr> <td>3</td><td>0.5</td><td>0.58</td></tr> <tr> <td>4</td><td>0.89</td><td>0.23</td></tr> <tr> <td>5</td><td>0.28</td><td>0.028</td></tr> <tr> <td>6</td><td>0.09</td><td>0.50</td></tr> <tr> <td>7</td><td>0.84</td><td>0.38</td></tr> </table>	Sample	P(C1/Sample)	P(C2/sample)	1	0.85	0.57	2	0.21	0.99	3	0.5	0.58	4	0.89	0.23	5	0.28	0.028	6	0.09	0.50	7	0.84	0.38	
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Q. 4	For the graph given below, frame the ensemble classifier. Derive the hypothesis equation on choosing $x < 5$ and $y < 3$ as classifiers. Apply them in order. Assume the points satisfying the classifier as star points else diamond. 	6 M (3+3)																								

***** ALL THE BEST *****